BTEC Level 3 Applied Science · Unit 1 Biology · Worksheet

Prokaryotic and Eukaryotic Cells

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

Learning focus: comparison tables, plasmids and medical applications.

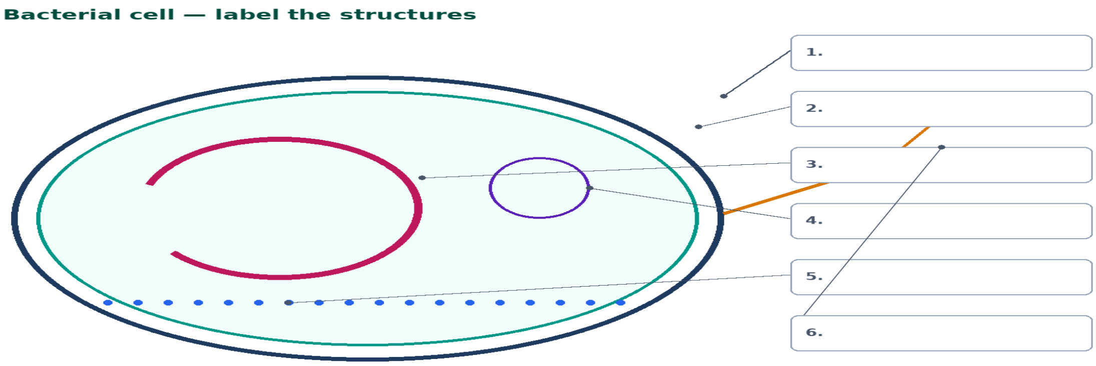
A Retrieval

1. What is the defining difference between prokaryotic and eukaryotic cells?

2. Name four structures of a typical bacterial cell.

3. Are viruses prokaryotic, eukaryotic, or neither? Justify.

4. Label the bacterial cell at the numbered leader lines: wall, membrane, DNA loop, plasmid, ribosomes and flagellum.



B Comparison table

5. Complete the comparison table.

Include nucleus, DNA form, organelles, ribosome type, typical size and wall chemistry.

Feature	Prokaryote	Eukaryote

6. What is a plasmid, and why can it be medically important?

7. Explain why some antibiotics can kill bacteria without killing human cells.

D Exam-style practice

8. Give three differences between a bacterial cell and an animal cell. [3]

9. Explain how plasmids can be medically important. [3]

10. Explain why bacteria are described as prokaryotic. [2]

E Challenge

11. Mitochondria and chloroplasts contain circular DNA and 70S ribosomes. Suggest how this observation is used in the endosymbiotic theory.